# Contents Lecture 10

- Linear programming
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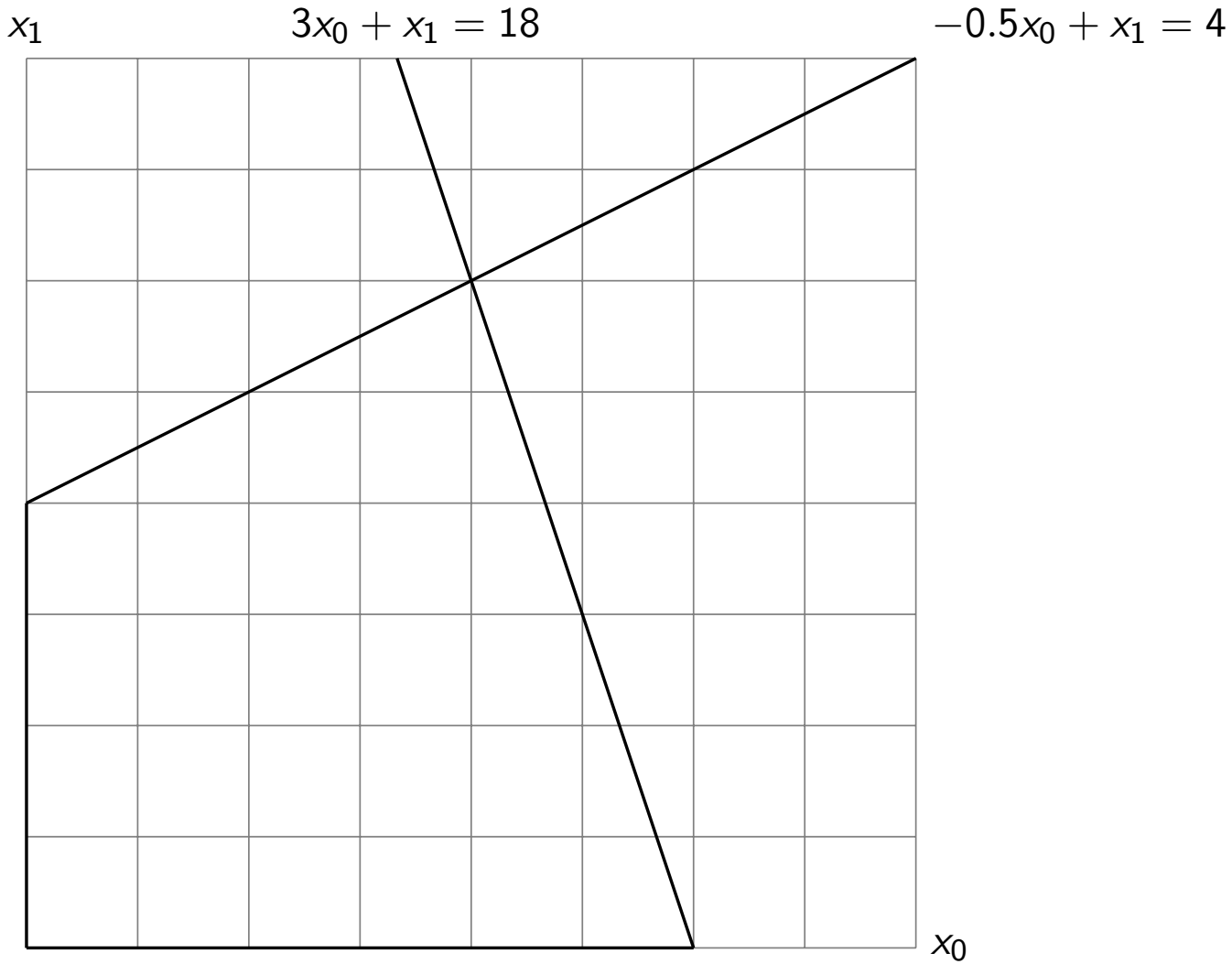
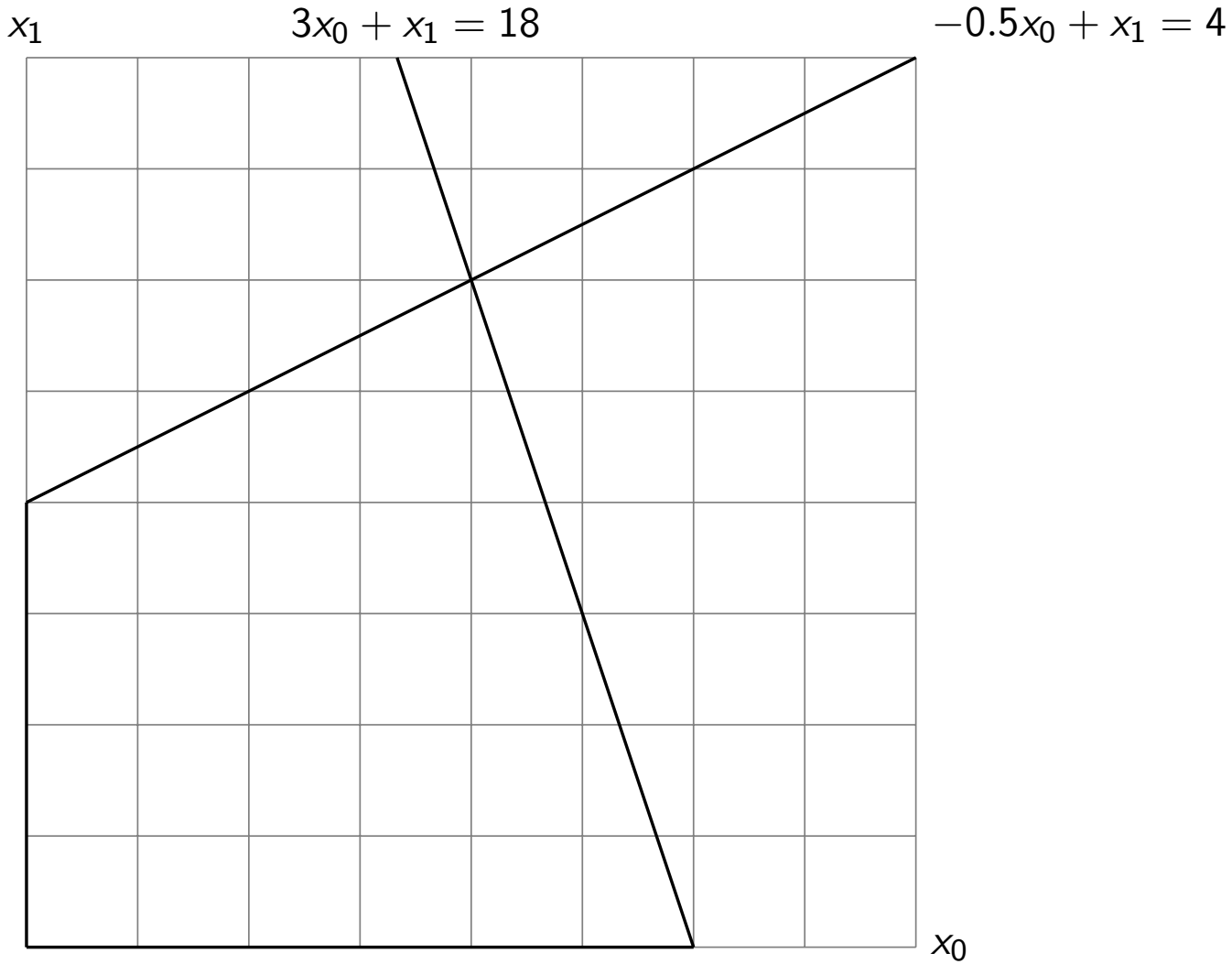
# Linear program: maximize a linear function in a region

- a region defined by lines including $x_i \geq 0$, i.e. linear constraints, and
- a linear objective function such as $\max z = x_0 + 2x_1$



# Solving a linear program

- Find $x_i \in \mathbb{R}$ which maximize $z$
- Here $x_i$ are called decision variables



# Our example

- Objective function and linear constraints using $\leq$

$$\max \quad z \quad = x_0 + 2x_1$$

$$\begin{array}{rcrcr} -0.5x_0 & + & x_1 & \leq & 4 \\ 3x_0 & + & x_1 & \leq & 18. \end{array}$$

- Also: implicitly $x_i \geq 0$
- We can also use $\geq$ and $=$ by first rewriting them to use $\leq$
- (with two $\leq$ instead of one $=$)

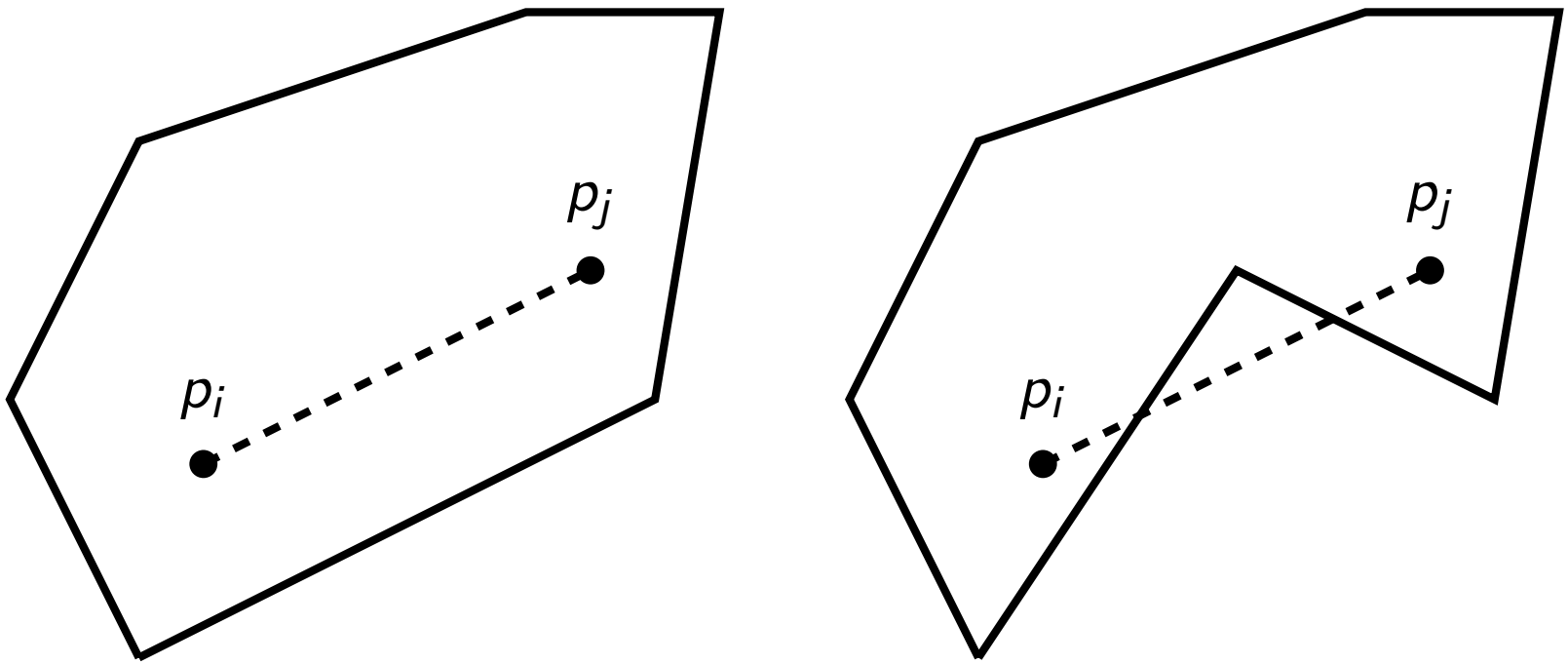
# Linear programs

- $$\begin{array}{lll} \max & z = c_0x_0 + c_1x_1 + \ldots + c_{n-1}x_{n-1} & \\ & a_{0,0}x_0 + a_{0,1}x_1 + \ldots + a_{0,n-1} & \leq \quad b_0 \\ & a_{1,0}x_0 + a_{1,1}x_1 + \ldots + a_{1,n-1} & \leq \quad b_1 \\ & \ldots & \\ & a_{m-1,0}x_0 + a_{m-1,1}x_1 + \ldots + a_{m-1,n-1} & \leq \quad b_{m-1} \\ & x_0, x_1, \ldots, x_{n-1} & \geq \quad 0 \end{array}$$
- or simpler as

$$\begin{array}{lll} \max & z = \mathrm{cx} & \\ & \mathrm{Ax} & \leq \quad \mathrm{b} \\ & \mathrm{x} & \geq \quad 0. \end{array}$$

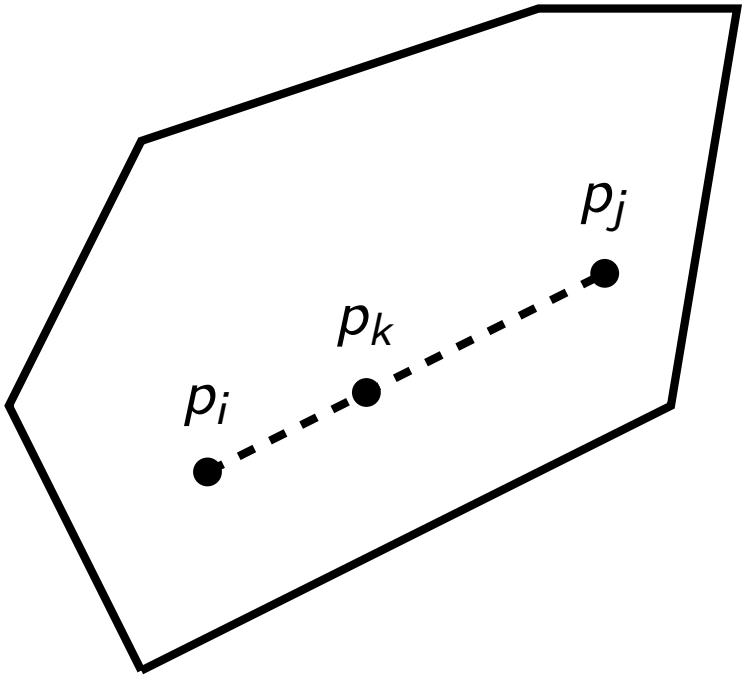
# Solutions

- Each constraint defines a halfplane in $n$ dimensions.
- The intersection of these halfplanes defines the **feasible region**, P, with **feasible solutions** $x \in P$.
- The feasible region is convex, and a point where halfplanes intersect is called a **vertex**.
- A non-convex region cannot be an intersection of halfplanes.



# Consequence of $P$ being a convex region

- Assume $p_k$ is on a line segment through $p_i$ and $p_j$
- We can write a point as $p_k = \lambda \cdot p_i + (1 - \lambda) \cdot p_j$, with $0 \leq \lambda \leq 1$
- So that $p_k$ is in the region
- Here $\lambda = 0.6$



# Solutions

- A linear program is either:
  - **infeasible** when P is empty,
  - **unbounded** when no finite solution exists, or
  - **feasible**, in which case we search for an optimal solution $x^* \in P$ which maximizes $z$.
- There may exist more than one optimal solution.

# Local and global optimal solutions

- We denote by $z(\mathrm{x})$ the value of the objective function $z$ at point x.
- A solution x is a **local optimum** for $z(\mathrm{x})$ if there is an $\epsilon > 0$ such that $z(\mathrm{x}) \geq z(\mathrm{y})$ for all $\mathrm{y} \in \mathrm{P}$ with $||x - y|| \leq \epsilon$.

**Theorem**

*A local optimum of a linear program is also a global optimum.*

**Theorem**

*For a bounded feasible linear program with feasible region $P$, at least one vertex is an optimal solution.*

- So we only need to check $z$ in the vertices and not the inner part of the region.

# A local optimum of a linear program is also a global optimum

**Proof.**

- $z(u)$ is a linear objective function
- Assume $z(u) > z(v)$ for all $v$ at most $\epsilon$ from $u$
- Let $w$ be any point in $P$, possibly far away from $u$ and $v$
- $v = \lambda u + (1 - \lambda)w$ with $0 \leq \lambda \leq 1$
- $z(u) \geq z(v) = z(\lambda u + (1 - \lambda)w) \geq \lambda z(u) + (1 - \lambda)z(w)$
- So $z(u) - \lambda z(u) \geq (1 - \lambda)z(w)$
- And $z(u)(1 - \lambda) \geq (1 - \lambda)z(w)$
- $0 \leq \lambda \leq 1$, so $z(u) \geq z(w)$

$\square$

# At least one vertex is an optimal solution

**Proof.**

- Let the feasible region $P$ have $k$ vertices: $\mathrm{x}^0, \mathrm{x}^1, ..., \mathrm{x}^{k-1}$
- Let $z^*$ be the maximum value in any vertex: $z^* = \max\{\mathrm{cx}^i, 0 \leq i < k\}$
- Every point $w$ in $P$ can be written as a linear combination of the vertices: $w = \sum_{i=0}^{k-1} \lambda_i \mathrm{x}^j$ with $\sum_{i=0}^{k-1} \lambda_i = 1$
- Let $w$ be any point in $P$
- $z(\mathrm{w}) = \mathrm{cw} = \mathrm{c} \sum \lambda_i \mathrm{x}^i = \sum \lambda_i (\mathrm{cx}^i) \leq \sum \lambda_i z^* = z^* \sum \lambda_i = z^*$.

$\square$

# Slack form

$$
\begin{aligned}
\max \quad & cx \\
x_{n+0} &= b_0 - \textstyle\sum_{j=0}^{n-1} a_{0,j} x_j \\
x_{n+1} &= b_1 - \textstyle\sum_{j=0}^{n-1} a_{1,j} x_j \\
&\dots \\
x_{n+m-1} &= b_{m-1} - \textstyle\sum_{j=0}^{n-1} a_{m-1,j} x_j \\
x_i &\geq 0 \qquad 0 \leq i \leq n+m-1
\end{aligned}
\tag{1}
$$

- The variables on the left hand side are called **basic variables** and occur only once, i.e. neither in any sum on the right hand side, nor in the objective function.
- The other variables are called **nonbasic variables**.

# Slack form of our example

- We start with

$$\begin{array}{lll} \max & z & = x_0 + 2x_1 \end{array}$$

$$\begin{array}{rcrcl} -0.5x_0 & + & x_1 & \leq & 4 \\ 3x_0 & + & x_1 & \leq & 18 \end{array}$$

- and then introduce two new variables, one for each constraint, and write it on slack form:

$$\begin{array}{lll} \max & z & = x_0 + 2x_1 + y \end{array}$$

$$\begin{array}{rcrcrcrcl} x_2 & = & 4 & - & ( & -0.5x_0 & + & x_1 & ) \\ x_3 & = & 18 & - & ( & 3x_0 & + & x_1 & ). \end{array}$$

- All $x_i \geq 0$ and $y$ is initially zero
- We rewrite the problem until all coefficients in the objective function become negative, and set all nonbasic variables to zero

# Entering and leaving basic variables

- Select a nonbasic variable with positive $c_i$ coefficient
- We take nonbasic variable $x_0$ as the so called **entering basic variable**

$$\max \quad z \quad = x_0 + 2x_1 + y$$

$$\begin{array}{rcrcrcrcr} x_2 & = & 4 & - & ( & -0.5x_0 & + & x_1 & ) \\ x_3 & = & 18 & - & ( & 3x_0 & + & x_1 & ). \end{array}$$

- Since $c_0$ is positive, we want to increase $x_0$ as much as possible
- The basic variables can limit how much $x_0$ may be increased (if there is no restriction, then the linear program is unbounded)
- $x_3$ restricts increasing $x_0$ to at most 6.
- Therefore we select $x_3$ as the so called **leaving basic variable**.

# Rewritten linear program

- We rewrite the linear program by letting the entering and leaving basic variables switch roles.
- This is a tedious but simple algebraic manipulation
- Do this by hand at least once

$$\max \quad z \quad = -0.333x_3 + 1.667x_1 + 6$$

$$\begin{array}{rcrcrcrcr} x_2 & = & 7 & - & ( & 0.167x_3 & + & 1.167x_1 & ) \\ x_0 & = & 6 & - & ( & 0.333x_3 & + & 0.333x_1 & ) \end{array}$$

- Next we must select $x_1$ as entering basic variable
- $x_2$ is restricted by $7 - 1.167x_1 \geq 0$
- $x_0$ is restricted by $6 - 0.333x_1 \geq 0$
- $x_2$ is most restricted and becomes the leaving basic variable

# Solution

- All $c_i$ are negative so $z$ cannot be increased with positive values of the nonbasic variables.
- By setting the nonbasic variables to zero the maximum becomes 16 in $x = (4, 6)$ which indeed is a vertex.

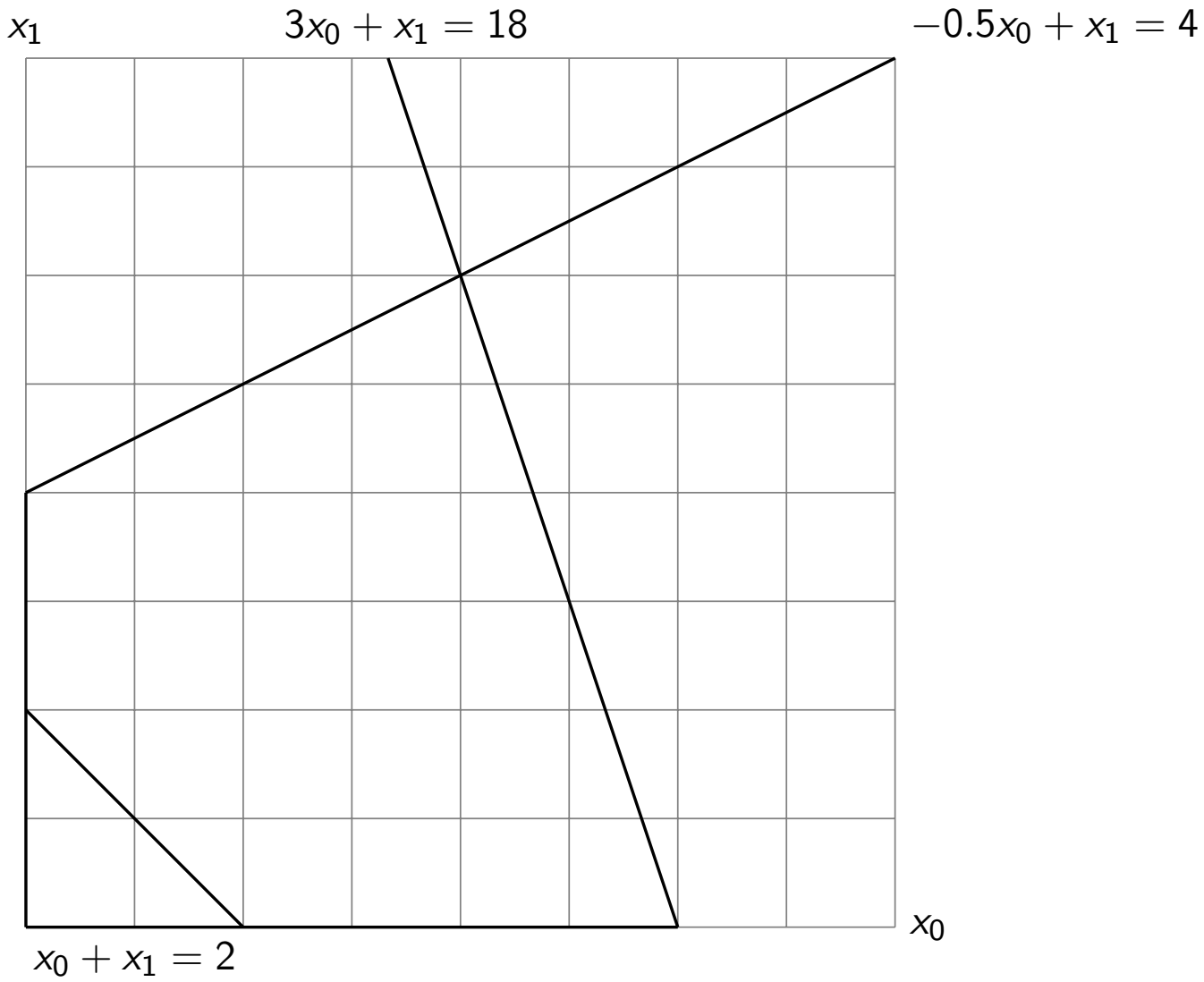
$$\max \quad z \quad = -0.6x_3 - 1.4x_2 + 16$$

$$\begin{array}{rcccccc} x_1 & = & 6 & - & ( & 0.1x_3 & + & 0.9x_2 & ) \\ x_0 & = & 4 & - & ( & 0.3x_3 & - & 0.3x_2 & ) \end{array}$$

- Summary: we start in a vertex and then go to a neighboring vertex until all coefficients are negative, which gives the optimal solution.
- It was an open problem but George Dantzig was late for a lecture at Berkeley and mistook it for a home assignment (he got a PhD for it).

# A problem

- We cannot start simplex in $(0, 0)$ if it is not in the feasible region



# Finding a start vertex when there is a negative $b_i$

- Let our original linear program be $P_0$
- If some $b_i$ is negative the point 0 is not in the feasible region
- We create a new problem $P_1$ to find a start vertex for $P_0$
- Add a new nonbasic variable $x_{n+m}$
- Start with $P_0$ and subtract $x_{n+m}$ from each constraint:

$$a_{i,0}x_0 + a_{i,1}x_1 + \ldots + a_{i,n-1} - x_{n+m} \quad \leq \quad b_i$$

- Use the objective function $z_1 = -x_{n+m}$
- We do a pivot on $P_1$ with $x_{m+n}$ as entering basic variable and $x_k$ with most negative $b_k$ as leaving basic variable
- This gives us b $\geq 0$, so $P_1$ can be solved using the simplex algorithm,
- If $P_1$ has optimal value 0 then $x_{n+m} = 0$ and by removing $x_{m+n}$ from this solution, we have a start vertex for $P_0$

# Integer programming

- Integer programming is similar to linear programming with the extra constraint that $x_i \in \mathbb{N}$.
- Some problems including this have no efficient algorithms
- One bad "method" to solve problems is to enumerate all solutions
- This does not sound good though
- We will use the algorithm design paradigm branch-and-bound to solve integer programs (not all due to integer programming is NP-complete)

# Branch

- A **relaxation** makes a problem simpler (by solving another problem)
- Perfekt tentasvar: "man softar problemet".
- For integer programming we solve the corresponding linear program, i.e. relaxing the integer requirement on the solution.
- Suppose we have an integer program and give it to the simplex algorithm and $x_k \notin \mathbb{Z}$
- Assume the simplex algorithm assigns $x_k = u$
- We can then branch by creating two new linear programs:
  - one with the additional constraint constraint $x_k \leq \lfloor u \rfloor$, and
  - another with the additional constraint $x_k \geq \lceil u \rceil$.
- Each new problem is solved directly with the simplex algorithm
- If it has an integer solution we can limit the search tree (bound)
- If it has a non-integer solution and it is better than the best integer solution we put it in a queue

# Bound

- If the simplex algorithm found an integer solution we do the following
- We check if this is the best integer solution found so far, and remember it in that case
- We remove from the queue all unexplored linear programs whose optimal value is less than the value of the integer solution we just found

# Graph coloring

- Consider an undirected graph $G(V, E)$ with $m$ edges and $n$ vertices.
- At most $n$ colors are needed and we create $n$ binary decision variables $w_i$ for $0 \leq i < n$, with the meaning $w_i = 0$ if color $i$ is unused and $w_i = 1$ if at least one vertex is assigned color $i$.
- For each vertex $v$ we then create $n$ decision variables $x_{vi}$, also binary, with the meaning that $x_{vi} = 1$ if vertex $v$ is assigned color $i$ and otherwise zero. The object function is to minimize the number of used colors, i.e., the sum of all $w_i$. There are two types of constraints:
  - each vertex must be assigned a color, so the sum of all $x_{vi}$ for a particular vertex $v$ must be one, and
  - two neighbors cannot use the same color, so if $u$ and $v$ are neighbors $x_{ui} + x_{vi} \leq w_i$, for all $i$.

# First model

The model becomes:

$$\min \quad z \quad = w_0 + w_1 + \ldots + w_{n-1}$$

$$\sum_{i=0}^{n-1} x_{vi} = 1 \qquad \forall v \in V$$

$$x_{ui} + x_{vi} \leq w_i \quad \forall (u, v) \in E$$

$$x_{vi}, w_i \in \{0, 1\} \quad \forall v \in V, 0 \leq i < n$$

# Second model

Since we expect the constraints to be on the form Ax $\leq$ b, we write it as follows instead:

$$\min \quad z \quad = w_0 + w_1 + ... + w_{n-1}$$

$$\begin{aligned}
\sum_{i=0}^{n-1} x_{vi} &\leq 1 \quad \forall v \in V \\
\sum_{i=0}^{n-1} -x_{vi} &\leq -1 \quad \forall v \in V \\
x_{ui} + x_{vi} - w_i &\leq 0 \quad \forall (u, v) \in E \\
x_{vi}, w_i &\leq 1 \quad \forall v \in V, 0 \leq i < n
\end{aligned}$$

# Instruction scheduling

- This is an important part of all optimizing compilers.
- The purpose is to minimize delays in the pipeline of a CPU. Assume for simplicity the input is a list of instructions, each with two source operands, one destination operand and a number indicating how many clock cycles it takes a CPU to execute the instruction.
- For instance, for a floating point add instruction, this **latency** may be five clock cycles. The input can be described as a weighted directed acyclic graph with nodes being instructions and with an edge from $u$ to $v$ if instruction $u$ computes the value of an operand of instruction $v$. The edge the latency of $u$ as weight.
- The instruction scheduling problem is NP-complete for most realistic CPUs. How can we solve it with integer linear programming? Let our goal be to find an $m$ cycle schedule, and that our CPU can issue ("start executing") $r$ instructions each clock cycle.

# Instruction scheduling

- Let $x_i^j$ be a decision variable which says instruction $i$ is scheduled in cycle $j$.
- Our schedule must satisfy the dependences in the dag, i.e., with an edge from $u$ to $v$, $u$ must not be scheduled after $v$ for correctness, but preferably sufficiently earlier than $v$ to take the latency of $u$ into account.
- Denote by $L_{uv}$ this latency. In this example we look for any feasible solution.
- With the nodes $V$, with $n = |V|$, and edges $E$, a basic model then becomes:

# Instruction scheduling model

$$\sum_{j=1}^{m} x_i^j = 1 \qquad \forall x_i \in V$$

$$\sum_{i=1}^{n} x_i^j \leq r \qquad \forall x_i 1 \leq j \leq n$$

$$\sum_{j=1}^{m} j \times x_k^j + L_{ki} \leq \sum_{j=1}^{m} j \times x_i^j \quad \forall (k,i) \in E$$

In the last constraint, the expression $\sum_{j=1}^{m} j \times x_k^j$ is the cycle in which instruction $k$ is scheduled.

# EDAG01 Efficient C, LP2

- This course has six labs with the labs 1-4 about implementing an ILP solver in C
- You can find files for the course in the tresorit-directory found at `cs.lth.se/edag01`
- Lab 5 is about understanding what takes time using a simulator from IBM Research
- Lab 6 is about what optimizing compilers can do
- There is a competition and the most recent winners are:
  - 2024: Karl Hedberg (D) and Alve Lindell (D and F)
  - 2023: Markus Bolinder (F)
  - 2022: Filip Raguz (N)
  - 2021: Erik Danielsson ($\pi$)